

DECODED EXPLAINED

1st September, 2026

AstroRad Vest Offers Lighter Shield Against Solar Storms



NASA astronaut evaluates the AstroRad vest aboard the International Space Station in 2022. (20042) (C.09.04)

AstroRad vest offers lighter shield against solar storms

Yasmeecan Mubumh

Despite their best efforts to settle the moon and later Mars, humans may never become a spacefaring species, more than a few experts have said over the years. An important reason for this view is radiation. The earth has a magnetic field that keeps harmful radiation at bay, but astronauts who fly beyond this field are at risk of developing cancer and other serious health problems.

A common fix for this problem has been to add more shielding for astronauts. However, every additional kilogram increases the launch cost by a large amount, and it is often also impractical to add enough shielding to a crew capsule to cope with a major outburst of radiation from the sun.

To this end, an Israeli company named Stenshal has, along with Lockheed Martin, developed a vest called AstroRad. Stenshal had originally made the suit for emergency responders on the ground.

Radiation remains a major obstacle to long duration human spaceflight because astronauts beyond earth's magnetic field face increased risks of cancer and other health problems

who may be exposed to nuclear radiation. The company adapted it for spaceflight, with a twist. AstroRad has more safeguards over the chest, abdomen, and pelvis, where there are large amounts of bone marrow, which make blood, and sensitive organs like the heart and the lungs. It also uses hydrogen-rich polymers to block radiation - materials rich in hydrogen slow energetic particles efficiently because hydrogen nuclei are similar in mass and so maximize energy transfer during collisions - rather than lead, which is more dense.

In 2019 and 2020, NASA arranged for astronauts to test AstroRad aboard the International Space Station. They wore it as they performed ordinary activities and gave feedback on its comfort and functionalities, allowing engineers on the ground to make improvements.

NASA tested the improved version onboard its Artemis I mission in 2022. In this mission, the crew capsule carried two human-shaped dummies named Helga and Zohar as it travelled around the moon before returning to the earth. Both Helga and Zohar were fit with materials designed to mimic human tissue as well as radiation sensors. Zohar wore AstroRad while Helga did not.

The results of analysing the data retrieved from the dummies were published in *Science Advances* on August 12. Using data collected during Artemis I to inform models of historical solar radiation events, researchers - from Stenshal, Lockheed, the German, Israeli, and U.S. space agencies, and institutions in the U.S. - found AstroRad could reduce the effective radiation dose an astronaut would have received during the August, 1972 solar storm by 60% and during the October 1989 solar storm by 40%. These amounts to 83 and 131 fewer days, respectively of exposure to deep-space radiation.

For the team, for astronauts heading back to the moon and eventually to Mars, AstroRad could allow astronauts to partially work through a solar storm instead of having to shelter themselves for the duration - a small but important step towards making interplanetary human spaceflight a safe reality.

That said, one particular shortcoming of the vest is that it can only deal with particles from the sun. Galactic cosmic rays that originate beyond the solar system remain harder to block.

AstroRad Vest Offers Lighter Shield Against Solar Storms

- **The Problem**

- Radiation remains a major obstacle to long-duration human spaceflight, since astronauts beyond Earth's magnetic field face increased risks of cancer and other health problems.
- A common fix adding more shielding increases launch costs sharply per kilogram, and is often impractical for a crew capsule to cope with a major solar radiation outburst.

- **The Solution**

- Israeli company StemRad, with Lockheed Martin, developed AstroRad, adapting a vest originally made for ground-based emergency responders exposed to nuclear radiation, adding shielding over the chest, abdomen, and pelvis areas rich in bone marrow and sensitive organs.
- AstroRad uses hydrogen-rich polymers rather than lead, since hydrogen-rich materials slow energetic particles more efficiently hydrogen nuclei are similar in mass to these particles, maximising energy transfer, unlike denser lead.

Why the US, World's Top Oil Producer, Wants to Control Venezuela's Crude

Why the US, world's top oil producer, wants to control Venezuela's crude

Sukalp Sharma

New Delhi, August 31

THE US says it will gain majority control over more than 65 billion barrels of Venezuelan oil — or 20% of its proven oil reserves — under an agreement with the South American country.

US President Donald Trump said this “biggest oil deal in world history” will be implemented through a partnership involving private companies “at no cost to the American taxpayer”.

Venezuela has the world's largest proven oil reserves, estimated at about 300 billion barrels, but produces and sells only about 1% of the crude the world is using. After the US captured Venezuela's President Nicolás Maduro in January, Trump declared that Washington would take control of the country's oil sector and that American majors would invest billions of dollars to revive it.

Now, his announcement represents an ambitious attempt to combine American capital and technology with Venezuela's

massive but dilapidated oil production infrastructure. The deal could mark the beginning of a reorientation of Venezuela towards the US.

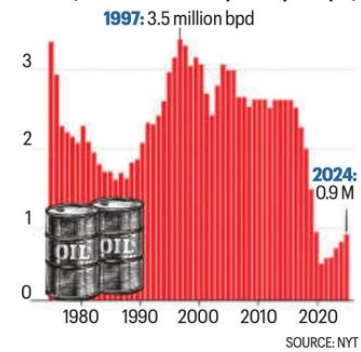
Globally, the US is the largest oil producer and also a top exporter. So, why does it want to have a high degree of control of Venezuelan crude?

Some reasons are self-evident — energy market dominance, reducing international oil prices amid the Strait of Hormuz crisis, and building up reserves. But a key reason is the very nature of Venezuelan oil — highly viscous, sticky, and “sour”.

Big announcement, few details

At this stage, the precise legal and commercial structure of the US-Venezuela agreement remains unclear. Some reports suggest the deal would likely involve a new private company jointly owned by US interests and an operator, with the US side holding a 55% operational share. The company would develop 17 oil fields, while US buyers would purchase oil at cost. The Venezuelan govern-

• Venezuela's oil production (in million barrels per day or bpd)



ment, meanwhile, has put the potential economic impact at roughly \$100 billion in private investment and more than \$209 billion in tax revenues.

Venezuela has historically maintained strong state control over its oil industry. A deal giving foreign companies significant ownership or operational control of its oil producing assets could become politically contentious.

Moreover, there are questions over whether Venezuela's interim administration has the authority to commit to long-term oil agreements. Reports indicate that opposition leaders in Venezuela are already up in arms against any deal giving control to American companies.

The legal structure of the new agreement will, therefore, be crucial.

Why US wants Venezuelan oil

Crude oil is not exactly a fungible commodity in practice because of the differences in quality of various types or grades of oil. And which type of crude is likely to be easily available in adequate volumes is a key factor in refinery configurations.

A key reason for US interest is the grade of Venezuela's oil, the bulk of which is clas-

sified as “heavy sour” crude: “heavy” because it is thicker and denser than the lighter crude oil grades, and “sour” because of its high sulphur content.

By contrast, the majority of US oil production is “light” and “sweet”, denoting low viscosity and sulphur content. But much of the American refining infrastructure — particularly the US Gulf Coast refineries — were specifically engineered to process heavy crudes from Latin America and Canada.

The US is also not on the best of terms with other countries that have significant heavy crude reserves, such as Iran and Russia. So, the US still has to import heavy sour crudes to feed these refineries, even as it exports massive volumes of its own “light sweet” oil.

There are also geopolitical considerations. The Trump administration wants a reliable and nearby supply source after its war on Iran prompted Tehran to choke oil supplies through the Strait of Hormuz — the narrow waterway that used to see a fifth

of the world's energy flows in peacetime.

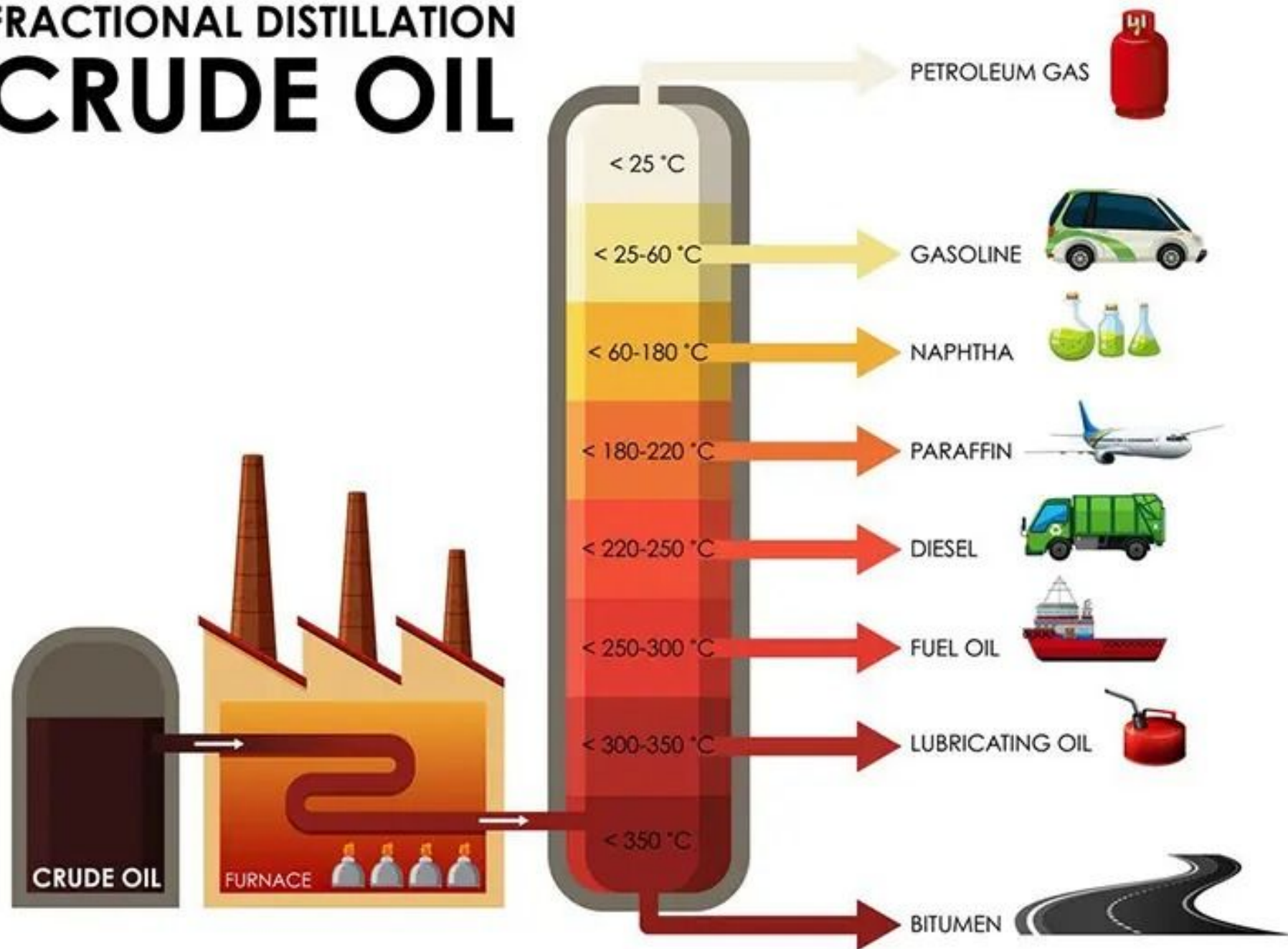
The challenges

Notwithstanding Trump's highly optimistic outlook for Venezuela's oil industry, scaling up production again will be no mean task. Norway-based energy research and intelligence firm Rystad Energy had estimated in January that Venezuelan oil production can return to late 1990s levels by 2040 if the new investment cycle starts as early as 2026. For this to happen, over \$180 billion would have to be invested over the next 15 years.

With US sanctions on Venezuela greatly constraining the flow of Venezuelan crude in the international oil market, China became its leading destination. Over the past couple of decades, both China and Russia have invested heavily in the Venezuelan petroleum industry.

Now, the outlook for these investments is uncertain. In that context, Trump's declaration can also be seen as Washington's attempt to cut Chinese and Russian influence in the Americas.

FRACTIONAL DISTILLATION CRUDE OIL



Renewed War Erases Russian Oil Discounts

Crude oil is a black liquid found in geological formations. It is a fossil fuel, which means it is formed from dead organisms that are buried under intense heat and pressure.

But not all crudes are alike. There are three primary qualities that differentiate one oil from another: Weight, Sweetness, and TAN count.

Weight of Crude: Light Oil vs Heavy Oil

Heavy oil: Evaporates slowly and Contains materials used to make products like asphalt

Light oil: Produces a greater percentage of gasoline and diesel and requires less processing

Based on Sulphur Content : Sweet or Sour Crude Oil

Sweet crude has very low levels of sulfur, well under 1%.

Sour crude has as much as 1-2% of sulfur.

Renewed War Erases Russian Oil Discounts

The three most well known regional benchmark oils are WTI, Brent, and OPEC.

- WTI stands for West Texas Intermediate. This is oil produced in the United States.
 - Typically lighter and Sweet
- Brent oil comes from the Scottish Brent and Ninian Systems located in the North Sea.
 - Typically lighter and Sweet
- OPEC stands for “Organization of Petroleum Exporting Countries.” It is a collective group of crude oils from several countries including Saudi Arabia, Iraq, Kuwait, Iran, Libya, Nigeria, Republic of the Congo, Equatorial Guinea, Gabon, and Venezuela.
 - Venezuela crude is among the heaviest and most sulfurous in the world
- Urals oil is a reference oil brand used as a basis for pricing of the Russian export oil mixture. It is a mix of heavy sour oil of the Urals and the Volga region with light oil of Western Siberia.

Why the US, World's Top Oil Producer, Wants to Control Venezuela's Crude

- **The Deal**

- The US says it will gain majority control over more than 65 billion barrels of Venezuelan oil (20% of its proven reserves), which Trump called the 'biggest oil deal in world history,' implemented through private companies 'at no cost to the American taxpayer.'

- **Why the US Wants Venezuelan Oil**

- Though the US is the world's largest oil producer, crude isn't fungible in practice, Venezuela holds the world's largest proven reserves (~300 billion barrels) but produces only about 1% of the crude the world uses, due to dilapidated infrastructure.
- A key reason: Venezuela's oil is mostly 'heavy sour' crude, which much of the US Gulf Coast refining infrastructure was specifically engineered to process, unlike the 'light sweet' crude that dominates US domestic production.

Renewed War Erases Russian Oil Discounts

The term 'West Texas Intermediate', sometimes found in news, refers to a grade of

This question was previously asked in

**UPSC Civil Services Exam
(Prelims) GS Official Paper-I...**

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1. Crude oil

2. Bullion

3. Rare earth elements

4. Uranium

Renewed War Erases Russian Oil Discounts

QUESTION

GS

Hard

Economy

Prelims 2011

In the context of global oil prices, “Brent crude oil” is frequently referred to in the news.

What does this term imply?

1. It is a major classification of crude oil.
2. It is sourced from the north sea.
3. It does not contain sulphur.

Which of the statements given above is/are correct?

[Select an option to attempt](#)

☐ A 2 only.

☐ B 1 and 2 only.

☐ C 1 and 3 only.

☐ D 1,2, and 3.

Navy Chief: INS Nipun Will Enhance Our Critical Underwater Capabilities

Navy Chief: *INS Nipun* will enhance our critical underwater capabilities

Amrita Nayak Dutta
New Delhi, August 31

INS NIPUN will significantly enhance the Navy's niche underwater capabilities required for multi-domain operations, Chief of Naval Staff Admiral Krishna Swaminathan said on Monday. He emphasised that this dedicated Diving Support and Submarine Rescue Vessel adds specialised capability for complex underwater operations and to respond to emergencies in one of the world's most "challenging and mysterious environments".

The Navy Chief was speaking at the commissioning ceremony of *INS Nipun* in Mumbai. Vice Admiral Sanjay Vatsayan, Flag Officer Commanding-in-Chief, Western Naval Command, other senior naval officers, representatives from Hindustan Shipyard Ltd., Visakhapatnam, among others, were present.

INS Nipun is the second Diving Support Vessel (DSV) of the Nistar Class and is a strong addition to the Navy's deep-sea diving and underwater support capabilities.

"We are living through a period of significant strategic



Navy Chief Admiral Krishna Swaminathan, Vice Admiral Sanjay Vatsayan along with other officers during the commissioning ceremony of *INS Nipun* in Mumbai on Monday. ANI

churn. The maritime environment is becoming increasingly contested, and the threat perceptions are constantly evolving and morphing into newer forms," Admiral Swaminathan said.

He noted that while it may not always be possible to predict the exact nature of the threats India will face at any given time, it is critical to stay prepared for anticipated threats and build robust capabilities to tackle these challenges effectively, wherever they may arise.

He said the addition of this

capability will enable the Western Naval Command to strengthen its capabilities to operate above, on, and beneath the waves.

Stating that the vessel brings together a range of capabilities possessed by only a handful of navies, which includes the ability to embark the Indian Navy's deep submergence rescue vessel, he said submarine rescue is a discipline where preparedness admits little margin for delay.

"*Nipun's* ability to embark and operate the Indian Navy's Deep Submergence Rescue

Vessel significantly enhances our capacity to respond to a distressed submarine quite significantly — whether our own, or that of a partner Navy. This capability would also enable India to support submarine rescue and emerge as the 'preferred submarine rescue partner' in the region," he said.

He added that the vessel's specialised diving capabilities can support partner nations in underwater salvage and HADR missions, adding another dimension to the Navy's maritime cooperation efforts.

He said the Western Naval Command will now have to integrate *INS Nipun* fully into operational plans, exercise her capabilities rigorously, develop proficiency around her systems, and ensure that the ship is ready at the earliest for every task at sea.

INS Nipun is capable of supporting saturation, air and heliox diving and is equipped with remotely operated underwater systems for inspection and intervention. Its specialised diving complex, dynamic positioning capability and associated life-support and medical facilities enable the conduct of sustained and complex operations at sea.

Navy Chief: INS Nipun Will Enhance Our Critical Underwater Capabilities

- **Nipun is a specialised platform designed to support a wide range of deep-sea diving and underwater operations, including diver support, underwater intervention, salvage missions, and complex maritime tasks.**
- **Nipun has been indigenously designed and constructed by Hindustan Shipyard Limited (HSL), Visakhapatnam,**

Will 'Talking' Cars Reduce Road Accident Deaths?

Will 'talking' cars reduce road accident deaths?

Dheeraj Mishra
New Delhi, August 31

THE UNION Ministry of Road Transport and Highways will soon mandate its new road-safety intervention, the Vehicle-to-Vehicle (V2V) communication system, through which cars will essentially talk with each other wirelessly to warn drivers about upcoming accident risks.

In the draft Central Motor Vehicles (Amendment) Rules, 2026, published on July 22, the Ministry had sought objections and suggestions from the public on the V2V system within 30 days.

Proposed rules

The draft Rules propose mandating the minimum technical and safety features for V2V-fitted vehicles before they are rolled out on roads, besides setting a deadline for manufacturers to install the system.

Vehicles such as two- and three-wheelers, cars, buses, and goods vehicles manufactured on or after October 1, 2027, will be required to comply with Automotive Industry Standard-230 (AIS-230), if fitted with a V2V system. Such vehicles, manufactured on or after October 1, 2028, will be required to be fitted with V2V, meeting AIS-230 specifications.

AIS-230 is a technical backbone for the V2V system, which specifies a list of requirements and provides for phased introduction of other safety features.

• AERIAL PARALLELS

- The V2V system is somewhat similar to technology used in aviation, where aircraft broadcast their position, speed, and altitude — and the nearby aircraft and ground station receive it.
- While this system is now common in the aviation sector globally, the road sector is still evolving.

How V2V works

For the V2V system, an On Board Unit (OBU) will be installed in vehicles so that it can exchange information with other vehicles wirelessly. It will use cellular Vehicle-to-Everything technology in the 5.875 GHz to 5.925 GHz frequency band — the Centre exempted its use from licensing requirements in June 2026. Now, the original equipment manufacturers can use this spectrum to get V2V installed in new vehicles.

The V2V-fitted vehicles will alert the driver about sudden braking, black spots, unsafe lane changes, obstacles like parked vehicles on roadsides, fog or any potential collision risk. For instance, if a car applies brakes suddenly, the nearby vehicles will get an alert to slow down before seeing it. Usually, V2V systems have a range of 300 metres and can detect vehicles in this range.

Officials said that OBUs will cost Rs 5,000 to Rs 7,000. This will be first installed in new vehicles.

The traditional safety features — such as the camera-based advanced driver assistance systems — depend on the line of sight of a driver, that is, how far one can see the road. The V2V system, however, works beyond the line of sight such as blind corners or behind large trucks, and it alerts the driver by establishing contact with other vehicles. So, it is expected to complement ADAS and predict forward collision risk. The vehicles will continuously exchange real-time data including speed, position, direction, and acceleration.

So far, V2V is working in a few countries such as the US.

Privacy concern

Since the V2V system will store a large amount of data about the vehicles, there is a potential risk of it being susceptible to cyber attacks.

It is unclear whether the allocated frequency band will support all vehicles. Also, miscommunication might lead to crashes.

Will 'Talking' Cars Reduce Road Accident Deaths?

- **Vehicle-to-Vehicle (V2V) technology is an advanced wireless communication framework that enables nearby motor vehicles to continuously broadcast and receive real-time data regarding their speed, location, direction, acceleration, and braking status.**
- **Unlike traditional sensors that require a direct line of sight, V2V enables proactive accident prevention by warning drivers about hidden hazards beyond their visual range.**
- **Voluntary Compliance Phase: From October 1, 2027, vehicles voluntarily equipped with V2V hardware must comply with AIS-230 rules.**
- **Mandatory Phase: Mandatory for all newly produced L, M, and N category vehicles manufactured on or after October 1, 2028.**

The Broken Promise of Right to Work

The broken promise of right to work

The replacement of MGNREGA with the VB-GRAM G Act has weakened the rural right-to-work framework, restoring a meaningful right to work through adequate funding, living wages and stronger local institutions would advance constitutional aims while also supporting rural demand and economic growth

ECONOMIC NOTES

Rajendran Narayanan
Vijay Ram S.

On August 21, a three-judge Supreme Court Bench led by the Chief Justice of India dismissed a petition and sought a fresh one in a case on minimum wages in rural employment guarantee programmes. The Bench lauded MGNREGA's role in rural development. It probed whether the 'Right to Work' should be treated on par with Article 21, the fundamental right to life, sparking an important discussion on the constitutional foundations of this right.

A guarantee under strain

Despite its limited scope of only 100 days of work per rural household and chronic underfunding, the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) was the only law anchored in the core principle of the right to work. In December 2025, the Union government replaced MGNREGA with the Viksit Bharat – Guarantee for Rozgar & Arogya Mission (Gramin) (VB-GRAM G) Act, and its implementation began on July 1. About 3.44 crore households worked for about 44 crore person-days under MGNREGA in July and August, on average, over the previous five years (Table 1). There has been a 60% decrease in employment in July and August this year compared with the average of the preceding five years. Notwithstanding the proverbial 'seething issues', the demand-driven right-to-work framework and the disproportionate fiscal burden that the new law places on States are likely to be major factors behind the dramatic fall in employment under the VB-GRAM G Act.

The impact of taking away even this limited right to work is therefore palpable. Discussions among the recent Supreme Court observations on the right to work can be traced back to the Constituent Assembly debates.

While most members agreed that a right to work was vital, the debate rested on whether it should be a fundamental right or be part of the Directive Principles of State Policy (DPSP). The DPSPs do not automatically constitute rights but are meant to serve as precepts for governments to formulate laws and policies. This became the basis of the differences in perspective.

Professor K.T. Shah had strongly advocated recognising the right to work as a fundamental right, arguing that the State needed a constitutional and positive legal mandate to guarantee socio-economic security to its citizens. Dr B.R. Ambedkar, among others, held that while the right to work was an essential goal, its immediate universal enforcement was not fiscally and institutionally viable in a newly independent India that had been hollowed out of its resources to deliberate constitutionalism of aspiration, rather than an abandonment of the welfare ideal. As scholars Anjan Kapur and Manu Bhargava have written in their essay 'Ambedkar and the anti-violation of Fellowship', "The constitution does not establish the nation, but rather provides the tools to help achieve it."

Therefore, Article 41 of the DPSP states: "The State shall, within the limits of its economic capacity and development,

A shrinking safety net

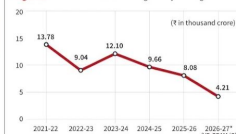
The shift from MGNREGA to VB-GRAM G has sharply reduced rural employment and earnings

Table 1: The number of households (HH) employed, person-days of work generated and average daily wages paid in July and August

Financial year	HH provided employment (in crore)	Person-days of work generated (in crore)	Average daily wage paid (in ₹)
2021-22	4.79	65.74	210
2022-23	3.13	40.14	225.5
2023-24	3.82	49.13	246.5
2024-25	3.04	37.36	259
2025-26	2.44	29.78	272
2026-27* (VB-GRAM G)	1.39	14.94	282.5

Sources: MGNREGA and VB-GRAM G MIS reports. *Data as on 31st August 2026 for FY 2026-27.

Figure 1: Estimated household earnings in July and August



*Data as on 31st Aug 2026 | Note: Total household earnings = sum of average wage rate paid x person-days of work generated in July and August. Sources: MGNREGA and VB-GRAM G MIS reports.



make effective provision for securing the right to work..." Embodying stronger progressive connotations, Articles 39, 42 and 43 call for the "right to an adequate means of livelihood, equal pay for equal work for both men and women, just and humane conditions of work" and to "secure to all workers a living wage and decent standard of life."

From aspiration to a statutory right

The Supreme Court, in *Odisha Panchayati Raj Union v. Panchayati Raj Union* (2023), ruled that the right to livelihood was a necessary condition for the fundamental right to life, giving a judicial fillip to the right to work. Spurred by the collective efforts of activists and rural workers, the National Rural Employment Guarantee Act was passed in 2005. Although limited in scope, it made a pan-India right-to-work law a reality for the first time. The State had to ensure its statutory obligation to provide employment at minimum wages.

In 2009, MGNREGA wages were de-linked from the Minimum Wages Act, 1948. Over time, MGNREGA wages barely kept pace with inflation and continued to remain lower than the minimum agricultural wages in most States. Low wages and an inadequate budget for generating employment have resulted in a steep decline in household earnings, with estimated total household earnings halving in July and August this year compared with the same months last year (Figure 1).

During the recent hearing, the Supreme Court Bench remarked that if financial resources remained the same, having a minimum wage threshold could reduce the number of workers who could

be provided employment. This remark, which appears to align with the Union government's approach to employment guarantee, merits scrutiny on at least two counts.

First, it vitates the judgment in *Sunil Roy v. State of Rajasthan* (1981) where the Supreme Court observed that payment of wages below minimum wages violates Article 23 of the Constitution and is akin to 'forced labour'. Second, it rests on a misleading economic logic that financial resources to implement a welfare programme must remain unchanged and

In reality, higher wages for rural workers increase their purchasing power, enhance their freedoms, and improve human development outcomes. These, in turn, increase effective demand for goods and services, creating a positive multiplier effect, resulting in higher economic productivity. Therefore, higher government spending on an employment guarantee programme at increased wages can yield a mutually reinforcing cycle between development and growth.

A retreat from the right to work by putting an arbitrary cap on funds for the VB-GRAM G Act and not linking it with minimum wages, the very notion of employment guarantee stands delegitimised. States are already grappling with fiscal constraints and curtailed borrowing limits under the Fiscal Responsibility and Budget Management (FRBM) framework. The increased burden of funding the VB-GRAM G Act will prevent States from generating employment. The sharp drop in employment and earnings is a reflection of this change.

THE GIST

The Supreme Court's recent observations have revived the Constituent Assembly-era debate over whether the right to work should be a fundamental right or remain under the directive principles.

Asking employment guarantee budgets constrained and wages below minimum wages can undermine both the constitutional promise of the right to work and rural household earnings.

Higher wages and greater public spending on employment can boost workers' purchasing power, stimulate demand for goods and services, and support wider economic growth.

States Seek Review of Wages, Funding Pattern for New Rural Job Scheme

Mahatma Gandhi National Rural Employment Guarantee Act

- MGNREGA is one of the largest work guarantee programmes in the world launched in **2005 by the Ministry of Rural development.**
- Rights based policy framework that guarantees 100 days of employment each year to rural households to undertake unskilled manual work
- It is a demand driven initiative
- Emphasis is on decentralised planning through Panchayat Raj Institutions. The act mandates Gram sabhas to recommend the works that are to be undertaken and at least 50% of the works must be executed by them.
- At least one-third of beneficiaries have to be women.
- Wages must be paid according to the statutory minimum wages specified for agricultural labourers in the state under the Minimum Wages Act, 1948.
- The employment will be provided within a radius of 5 km. If it is >5 km, an extra wage will be paid.
- Social Audit must be done by the Gram Sabha.
- **100% of the unskilled labour cost and 75% of the material cost (including wages of skilled and semi-skilled workers) of the programme is borne by the Centre.**
- State provides the unemployment allowance and compensation

States Seek Review of Wages, Funding Pattern for New Rural Job Scheme

Viksit Bharat - Guarantee for Rozgar and Aajeevika Mission, 2025

Ministry : Agriculture and Family Welfare

Provisions

- The Viksit Bharat Guarantee for Rozgar and Aajeevika Mission Gramin is the new rural employment and livelihood scheme that replaced the Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS) and will come into force from 1st July 2026.
- The scheme provides 125 days of guaranteed wage employment in a financial year to every rural household whose adult members volunteer to undertake unskilled manual work.
- It also mandates timely wage payments within 15 days of completion of work.
- The funding structure is designed according to state capacity, with a 60:40 Centre-State ratio, 90:10 for Northeastern and Himalayan states, and full (100%) central funding for Union Territories without legislatures.
- State governments can notify a combined 60-day no-work period during peak sowing and harvesting seasons to prevent agricultural labour shortages while ensuring workers still receive 125 days of employment during the remaining part of the year.
- The Act mandates preparation of Viksit Gram Panchayat Plans using spatial technologies such as the Yuktdhara portal and integration with national planning platforms like PM Gati Shakti.

States Seek Review of Wages, Funding Pattern for New Rural Job Scheme

MGNREGA

- 100 days of wage employment per rural household
- Multiple and scattered categories of works with limited strategic focus
- Center bears unskilled wage costs, states bear unemployment allowance
- No explicit statutory "pause window"
- Demand based funding with unpredictable allocations
- Gram Panchayat planning is central

Viksit Bharat-G RAM G

- 125 days of wage employment per rural household
- 4 clearly defined priority areas focusing on water security, rural infrastructure, livelihoods and climate resilience (Temporary assets to durable assets)
- State cost-sharing for wages, 60:40 for most states, 90:10 for certain special-category regions
- States can notify up to 60 days in a FY when work will not be executed
- Normative funding ensuring predictable budgeting while protecting the employment guarantee. Extra cost will be borne by state government (supply driven)
- Integrates institutionalised convergence and infrastructure planning

States Seek Review of Wages, Funding Pattern for New Rural Job Scheme

Q. Among the following who are eligible to benefit from the “Mahatma Gandhi National Rural Employment Guarantee Act”? (2011)

- (a)** Adult members of only the scheduled caste and scheduled tribe households
- (b)** Adult members of below poverty line (BPL) households
- (c)** Adult members of households of all backward communities
- (d)** Adult members of any household

V-Dem's Democracy Index Rates India at Its Lowest Since 1975

V-dem's democracy index rates India at its lowest since 1975

The UN CERD's concerns about the exclusion of a large number of voters is in line with the V-Dem Institute's data on India's democratic backslide

DATA POINT

Sambavi Parthasarathy
Gauri Singavarapu

Last week, the UN Committee on the Elimination of Racial Discrimination (CERD), said it was "gravely concerned" about "reports of large-scale violations perpetrated by law enforcement officials" in its first review of India since 2007. The committee's observations included concerns over the exclusion of large number of voters, Muslims in particular, as part of the Special Intensive Revision of electoral rolls.

UN CERD's criticism seems to be in line with some of the observations in the latest democracy report by the University of Gothenburg-based Varieties of Democracy (V-Dem) Institute, which uses a multidimensional approach to "conceptualising and measuring democracy." It noted India as one of the countries that is "autocratising" slowly with a systematic dismantling of democratic institutions. This is reflected in the institute's electoral democracy index released in March. While India's performance has gradually declined since 2009, the latest index score is India's lowest since 1975, when the country was under Emergency rule (Chart 1). India's scores across parameters such as judicial accountability and corruption, autonomy of the Election Management Body (EMB), and impartial administration by public officials are at their lowest in 50 years. (Charts 2, 3, 4).

The decline in autonomy of institutions, as per the report, seems to have accompanied deteriorations in freedom of academic and cultural expression and harassment of journalists (Chart 5). Academic freedom measured through indicators such as freedom of academic expression, freedom to research and teach, and institutional autonomy have also declined (Chart 6).

New depths

The data for the charts were sourced from the Varieties of Democracy (V-Dem) Institute, University of Gothenburg, Sweden.
Gauri Singavarapu is interning with The Hindu's Data team

CHART 1: V-Dem's democracy index score for India is at its lowest since 1975. It was 0.39 in 1975, slightly higher than the 0.38 recorded in 2025

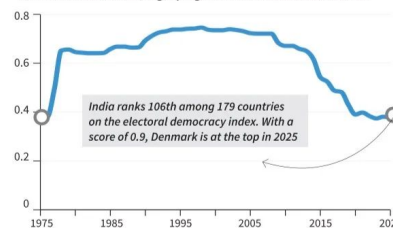


CHART 2: India's scores across election-related indicators involving the functioning of Election Management Body (EMB) and conduct of elections

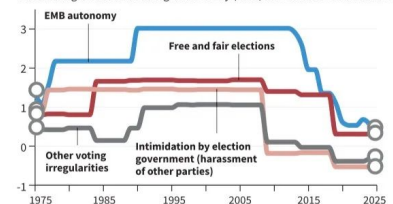


CHART 3: India's scores across indicators on religion, private property, access to justice, and public administration, as per V-Dem Institute's report

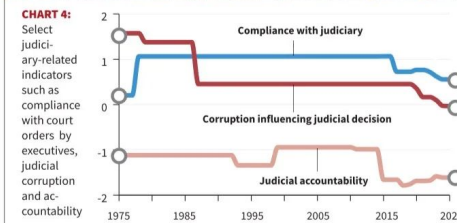
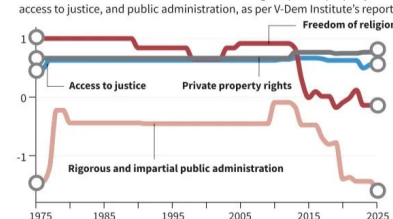


CHART 5: India's scores across indicators related to media and freedom of expression, as per V-Dem Institute's report

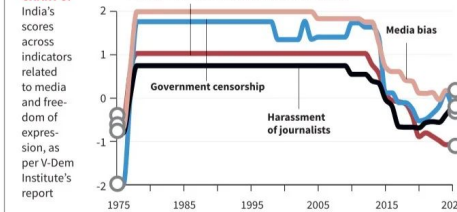
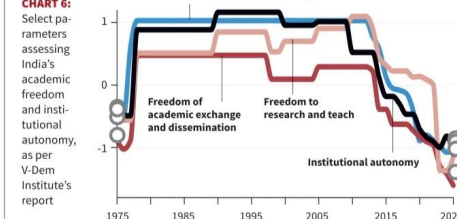


CHART 6: India's scores across indicators on academic freedom and institutional autonomy, as per V-Dem Institute's report



V-Dem's Democracy Index Rates India at Its Lowest Since 1975

- **The Varieties of Democracy (V-Dem) Index is a global framework that measures democracy using a multidimensional approach, rather than treating democracy merely as the conduct of elections.**
- **It evaluates countries across dimensions such as electoral integrity, civil liberties, institutional autonomy, judicial accountability, freedom of expression and public administration.**
- **Published by the Varieties of Democracy (V-Dem) Institute, based at the University of Gothenburg, Sweden.**

Deadly Span: India's Power Infrastructure Threatens Vultures

Deadly span

India's power infrastructure is emerging as a big threat to vultures

Whereas for many decades the chief concern about vulture safety was centered on chemical threats, the threat of electrocution posed by India's expanding power infrastructure may supersede it. In the 1990s, India's vultures suffered one of the world's worst population crashes, nosediving 99.5% by 2007 from a high of around four crore in the 1980s; the white-rumped, Indian, and slender-billed vultures were nearly wiped out. The culprit was diclofenac, a painkiller for cattle that wrecks vulture kidneys. The government banned the use of diclofenac, with additional bans in 2023 to mitigate threats posed by aceclofenac and ketoprofen, among some other NSAIDs. The birds' decline left carcasses exposed for longer, supporting feral dog populations and leading to rabies outbreaks; a 2024 study in American Economic Review estimated that human mortality thus rose 4% and the associated damages cost India \$69.4 billion a year. While electrocution has not become as deadly as NSAIDs have been, the fragility of the vulture population and the proliferation of unsafe power infrastructure warrant early action. One official survey reported in 2025 that vultures were nesting at only 50% of their historic nesting sites. It is no trivial detail that the raptors' wide wingspans mean a large individual can contact two conductors at once. They also have a habit of perching on elevated structures, are attracted to open landscapes, and have incentives to congregate where the availability of food is predictable, a risk enhanced by people dumping food waste around electrical installations.

As *The Hindu* reported based on a recent study in the *Journal of Threatened Taxa*, "relocating a livestock carcass dump 2.4 km away from high-tension power infrastructure near Uttarakhand's Dehradun may have saved these large scavengers from ... electrocution". An assessment prepared for the State Climate Resilient Power System Development Project also noted an Egyptian vulture and some steppe eagles had been electrocuted on medium-voltage, rather than high-voltage, lines. Research has noted that persistent mortality due to electrocution alone could render a local population extinct. Avian electrocution is also likely under-documented in India since dead birds can be removed by humans or scavenged. As the study indicated, one inexpensive solution is to separate vulture feeding sites from power infrastructure. Other measures to protect raptors, and even other birds, especially near settlements, include insulating conductors, increasing the gap between energised and grounded components or otherwise preventing a bird from bridging them – non-trivial considering vulture wingspans – and installing safe perches.

Thank You